

Contrastive Learning for Prediction of Alzheimer's Disease Using Brain 18F-FDG PET

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Abstract—Brain 18F-FDG PET images are commonly-known materials for effectively predicting Alzheimer's disease (AD). However, the data volume of PET is usually insufficient, which is unfavorable to train an accurate AD prediction networks. Furthermore, the PET image is noisy with low signal-to-noise ratio, and simultaneously the feature (metabolic abnormality) used for predicting AD in PET image is not always obvious. Therefore, a contrastive-based learning method is proposed to address the challenges of PET image inherently possessed. Firstly, the slices of 3D PET image are amplified by cropping the image of anchors (i.e., an augmented version of the same image) to generate extended training data. Meanwhile, contrastive loss is adopted to enlarge inter-class feature distances and reduce intra-class feature differences using subject fuzzy labels as supervised information. Secondly, we construct a double convolutional hybrid attention module to enhance the network to learn different perceptual domains where two convolutional layers with different convolutional kernels (7×7 and 5×5) are constructed. Moreover, we recommend a diagnosis mechanism by analyzing the consistency of predicted result for PET slices alone with clinical neuropsychological assessment to achieve a better AD diagnosis. The experimental results show that the proposed method outperforms the state-of-the-arts for brain 18F-FDG PET images, and hence demonstrate the advantage of the method in effectively predicting AD.

Index Terms—Alzheimer's disease (AD), deep learning, brain 18F-FDG PET, contrastive loss, multiattention mechanism, multi-correlation analysis.

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I. INTRODUCTION

ALZHEIMER'S disease (AD) is a neurodegenerative disease that leads to dementia [1] and is caused by the misfolding and chemical reaction of the $A\beta$ protein in the brain, resulting in neuronal inactivation [2] and brain tissue necrosis that threatens human life.

At present, the Mini Mental State Examination (MMSE) and Clinical Dementia Rating (CDR) are commonly used clinical neuropsychological assessment methods [3]. The MMSE and CDR not only help clinicians assess the severity of dementia in patients, but are also easily accepted by patients. However, clinical neuropsychological assessment is only a clinical assessment method, and it cannot diagnose the disease from the analysis of the manifest structure, secretory metabolism and lesion. Therefore, the early diagnosis of AD is not sufficient only based on the diagnosis of clinical neuropsychological assessment results.

With the rapid development of neuroimaging technology, neuroimaging has become an intuitive and reliable way to diagnose Alzheimer's disease [4]. In neuroimaging diagnosis, brain 18F-FDG PET is an important neuroimaging technique for detecting Alzheimer's disease. 18F-FDG is known as fluorine-18 fluorodeoxyglucose which is a radiotracer that can be used with PET imaging, which is an imaging that reflects the genetic, molecular, metabolic and functional status of brain lesions. When 18F-FDG PET is used to image the brain, it reflects glucose consumption in the active areas of glutamatergic synapses and astrocytes in the brain [5], and is a "barometer" of brain activity and neuronal function. Studies have shown that brain 18F-FDG PET shows features of AD neuropathy earlier than MRI of the brain in individuals with mild cognitive impairment [6]. Therefore, brain 18F-FDG PET neuroimaging is an important information for early screening of AD.

Given the excellent performance of 18F-FDG PET for AD, Liu et al. [7] proposed an AD risk prediction method based on brain 18F-FDG PET, using deep learning network to learn brain 18F-FDG PET slice information. The paper compared the prediction effect under different convolutional layers, and the result of taking High-order convolutional layer was classified in AD vs. MCI (mild cognitive impairment) and obtained more than 90% prediction accuracy. At the same time, Baydargi et al. [8] proposed a parallel network model for Alzheimer's disease risk prediction, and their proposed prediction model could reach 95.51% accuracy in AD vs. NC (normal controls) classification.

However, with the extensive experimental and research work carried out in the field of AD prediction, it has been found that these methods, although effective in predicting AD, use parametric loss functions [9] that can lead to low model robustness, poor margin of difference (margin that allows room for error) and low

recognition rate for brain 18F-FDG PET slices that are mislabeled, noisy, and have a single amount of information and severe artifacts in the training process. Therefore, we are eager for a loss function that enables the network to acquire higher-level generic feature relationships between images and take advantage of the inter-class differences between brain 18F-FDG PET images to learn features between classes.

Supervised contrastive loss [10] construct “positive and negative” samples through different forms of the same image and itself, guiding the network to close the distance between the positive sample features and the positive sample features, and pushing away the distance between the negative sample features [11], and reinforcing the learning of the features of the inter-class samples by measuring the similarity between different samples.

In the classification task of natural images, Grill et al. [12] proposed self-supervised contrastive learning achieved an accuracy of 79.16% under the ResNet structure. Khosla et al. [13] proposed supervised contrastive learning achieves an accuracy of 81.4% under the ResNet structure. Therefore, applying contrastive learning methods to early prediction studies of AD is a promising direction.

Based on these existing studies mentioned above, we can surmise that:

1. the issues of brain 18F-FDG PET slice images can lead to the poor classification performance of the traditional deep learning method. Meanwhile, the presence of lesions on some of the slices of the subjects do not match the label information, and that can reduce the network feature learning ability.
2. the traditional convolutional neural network cannot obtain feature information for brain 18F-FDG PET slice images with low signal-to-noise ratio and high noise.
3. the single use of clinical neuropsychological scale assessment is poor for AD prediction and cannot provide reliable information for clinical AD prediction.

To address the above problems, this paper proposes a contrastive learning-based method for AD prediction. The main contributions of this paper are as follows:

- We quantitatively augmented PET slices with filtering and cropping as extended data for contrastive anchors, while subjects' fuzzy labels were used as supervised information to expand inter-class feature distances and narrow intra-class feature differences, and to construct contrastive losses.
- We constructed a prediction network based on a double convolution hybrid attention mechanism in order to increase the feature-aware field range of the network while suppressing redundant information from PET slices, and visualized the network prediction process, and the results showed that the region of interest of the prediction network was consistent with the results of the AD lesion region in clinical studies.
- We further improved the accuracy of AD prediction by performing multi-correlation analysis on the predicted results of PET slices of the subjects and combined with the results of clinical neuropsychological assessment for comprehensive AD prediction.

The rest of this paper is organized as follows. In Section II, we present the related work of this paper. In Section III, we introduce the studied dataset and preprocessing method.

In Section IV, we propose a neuroimaging prediction network framework: Improved anchor augmentation approach, fuzzy supervised contrastive loss, double convolution hybrid attention mechanism, and correlation analysis methods. In Section V, we comprehensively evaluate the performance of the proposed method from a large number of experiments and analyses. The paper concludes with a summary and outlook in Section VI.

II. RELATED WORK

Early prediction of AD based on brain 18F-FDG PET can be simplified to a problem of image understanding. Li and Tang learn features from natural images with rich information, which makes it possible to fit multiple image understanding tasks [9]. Although all these methods have achieved excellent results in the field of natural images, they are not applicable in the field of neuroimaging. This is attributed to the complex nature and heterogeneity of neural images. To address these issues, we consider a better contrastive learning method for representation learning to obtain neuroimaging prediction model training.

The core idea of contrastive learning [14] is to close the distance between a certain anchor point and its positive samples and to increase the distance between the anchor point and the negative samples of that anchor point, there is only one positive anchor point sample, and all others are considered negative samples. In contrast, supervised contrastive learning [13] generates more positive samples from more anchors using label information, increases the inner product operation, and enhances intersample contrast. Additionally, lesion features in the subject's slice that do not match the label are present. We blurred the labels to reduce the rigidity of the supervised information. However, the slice images of brain 18F-FDG PET are noisy, and the noise type is undeterminable. Therefore, in this paper, we generated 2-fold data as training data set, which is augmented with positive and negative sample examples (such as image cropping, flipping and image blurring, image gaussian blurring, image median blurring, image bilateral blurring) to facilitate lesion feature learning by the prediction network.

The image attention mechanism initially mimics human thinking patterns and habits, aiming to read focused information from global images quickly and discard the nonfocused information in the images. In Woo et al. [15], a convolutional block attention module (CBAM) was proposed to infer an attention map along the channel and spatial dimensions, thus enhancing the representation of cnn. Han et al. [16] proposed an attribute-aware attention model that uses attribute information to select category features in different regions. In general, visual attention mechanisms have the advantage of discriminating the content and location of attention images for better category identification. Suk et al. [17] developed a model based on autoencoder (AE) networks for AD diagnosis and used several complex support vector machine kernels for classification. In this paper, we proposed a double convolution hybrid attention mechanism prediction network that incorporates a dual convolutional module that allows the prediction network to focus on more regions, to learn more reasonable features and reduces learning failure due to subject heterogeneity. we constructed a hybrid attention module (HAM) that alleviates the focus on redundant information in network images and enhances network learning of valid features.

Usually, the prediction of AD based on neuroimaging prediction results alone is often incomplete because neuroimaging prediction results are limited to subjects with abnormal brain

TABLE I
CHARACTERISTICS OF THE ADNI-1 AND ADNI-2 DATASET SUBJECTS USED IN THIS STUDY

Data sets	Category label	Case (n,M/F)	Age (year,Mean $\pm$ SD)	MMSE (Mean $\pm$ SD)	CDR (Mean $\pm$ SD)
Train/Val ADNI-1	AD	100/90	75.3 $\pm$ 7.5	23.3 $\pm$ 2.0	0.8 $\pm$ 0.25
	MCI	150/100	74.9 $\pm$ 7.6	27.3 $\pm$ 1.8	0.5 $\pm$ 0.03
	NC	100/100	75.8 $\pm$ 5.0	29.1 $\pm$ 1.0	0 $\pm$ 0
Test ADNI-2	AD	70/54	74.2 $\pm$ 8.0	23.2 $\pm$ 2.2	0.9 $\pm$ 0.25
	MCI	100/92	71.7 $\pm$ 7.6	28.3 $\pm$ 1.6	0.4 $\pm$ 0.03
	NC	50/34	74.8 $\pm$ 6.8	26.6 $\pm$ 1.7	0.01 $\pm$ 0

metabolism and may be less optimal for the classification of NC, MCI and AD with cognitive abnormalities [18], [19], leading to a clinical misdiagnosis phenomenon. The AD prediction methods of Lee et al. [20] and Zhou et al. [21] combined clinical information and achieved good prediction results. In this paper, the prediction results from PET slices from subjects were subjected to multi-correlation analysis and these results were combined with the results of clinical neuropsychological scales for comprehensive early screening to provide accurate information for treatment of clinical AD [22].

III. DATASETS

In this section, the dataset source and acquisition process are described in detail to demonstrate the preprocessing approach of this paper.

A. Data Collection and Acquisition

Due to the specificity of early AD prediction studies, the study in this paper was based on the ADNI public database. Two datasets consisting of 1040 subjects were used in this study, including 1) the ADNI-1 dataset and 2) the ADNI-2 dataset. The distribution of the subjects in the dataset is shown in Table I.

B. Data Preprocessing

To capture the features of brain 18F-FDG PET, 3D networks are usually constructed to learn spatial features and make predictions. Although 3D networks have been used for capturing neural image features and good results have been achieved [23], building a 3D network requires more parameters than building a 2D network and utilizes more training data to avoid overfitting in the training phase. Unlike in the natural image classification task, collecting sufficient neuroimaging data remains a challenge in clinical neuroimaging disease diagnosis applications due to the large amount of redundant information contained in PET data and the lack of complete lesion information. Therefore, it is not possible to construct a 3D prediction model with excellent classification performance.

To reduce the amount of brain 18F-FDG PET data and save computing costs, the 3D brain PET data were decomposed into 2D slices to train the model. Firstly, the PET data were normalized to an international brain mapping consortium template through the statistical parameter mapping (SPM12) toolbox. Secondly, averaging, spatial alignment, interpolation to a standard voxel size, intensity normalization, and other techniques were performed, and the rest of the parameters were set to their default values, the size of each normalized image was $79 \times 95 \times 79$. Finally, the PET results were sliced and tiled into

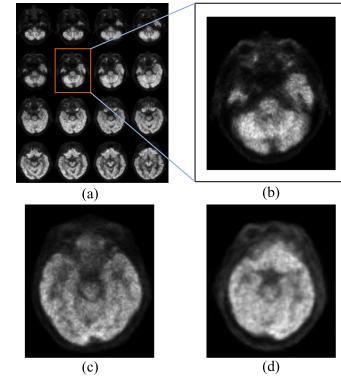


Fig. 1. Brain 18F-FDG PET slices of subjects after pre-processing.

a 2D image sequence via a data processing library (nibabel) in Python and then adjusted to 95×79 pixels.

Fig. 1 shows examples of fluorine-18 fluorodeoxyglucose PET images from the Alzheimer's Disease Neuroimaging Initiative set preprocessed with the proposed method for patients with AD. One representative zoomed-in section is provided for each of the three example patients: Fig. 1(b) shows a 76-year-old man with AD. Fig. 1(c) shows an 83-year-old woman with MCI, and Fig. 1(d) shows an 80-year-old man with NC. In this example, the patient with AD presented slightly less gray matter than did the patient with NC.

IV. METHODS

The neuroimaging prediction network consists of four key components: A backbone network and learning strategy, anchor points and a loss function, a double convolution hybrid attention module, and multi-correlation analysis with joint diagnosis. In this section, these four key components are introduced in detail, as shown in Fig. 2.

The training process of the neuroimaging prediction network consists of two stages. The first stage involves the improvement of the anchor augmentation method for brain 18F-FDG PET slices and the acquisition of the representation weights of the model for these slices by using a fuzzy supervised contrastive loss. In the second stage, the representation weights trained in the first stage are fixed, and then the slices are fed into the model to train the classifier via the cross-entropy loss to obtain the weights of the classifier.

The testing process also consists of two stages. Brain 18F-FDG PET slices are first fed into the backbone network and then predicted by the classifier in the second stage (the first stage developed at the time of training is completely discarded at this point). In the second stage, the consistency of the neuroimaging predictions is judged through correlation analysis, and a joint diagnosis is made by combining the results of the clinical neuropsychological assessment.

A. Backbone Network and Learning Strategy

The backbone network shown in Fig. 2 consists of the following components: 2 convolutional modules with a hybrid attention module (HAM), 2 maximum pooling modules and 16 residual units (L1-L4). To be specific, the convolutional kernel sizes of the two convolutions are (7×7) and (5×5) , respectively. Stage 1 in Fig. 2 is the process to learn the representation

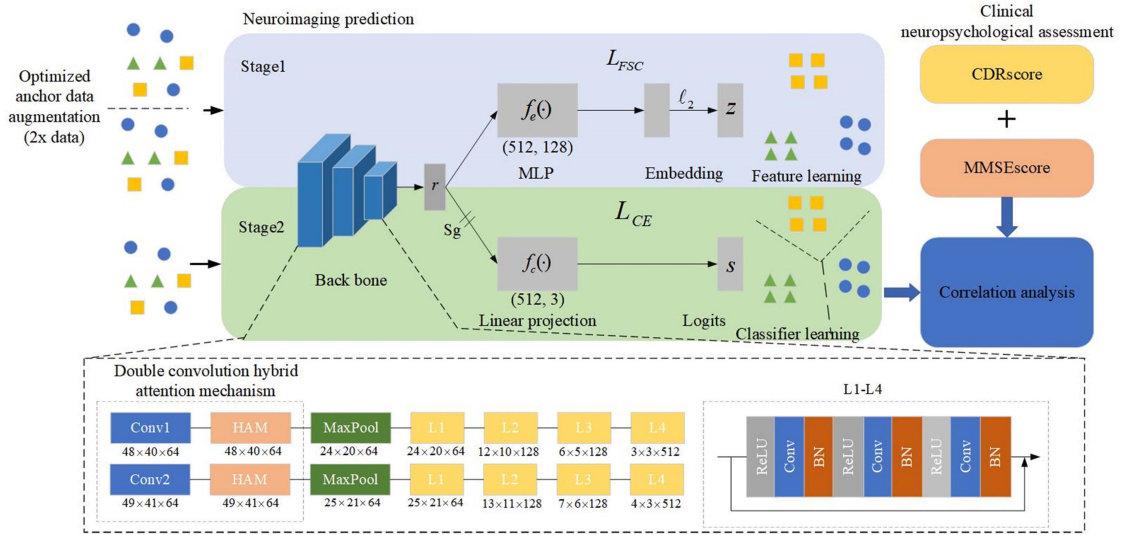


Fig. 2. Framework for AD diagnosis based on neuroimaging prediction and clinical neuropsychological assessment.

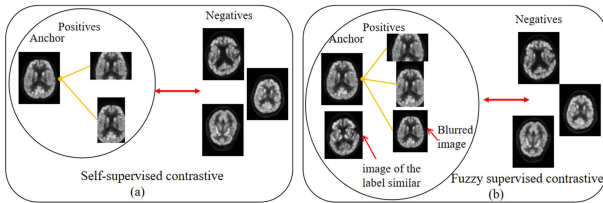


Fig. 3. Self-supervised contrastive vs. Fuzzy supervised contrastive. Adopted from [13].

of PET slices images. Such process follows a 2-steps pipeline: Step 1, the lesion features are extracted from PET slice images using the backbone. Step 2, the extracted lesion features, also known as the tensor information $r \in R^{D_E}$, $r \in R^{D_E}$ is fed into the projection head $f_e(\cdot)$, which is a nonlinear multilayer perceptron (MLP). Finally, the output of the pipeline is normalized to be a vector $z \in R^{D_S}$ using ℓ_2 norm, with output nodes parameters is 128 [13]. The norm is applied to z such that inner product can be used as distance measurements. Stage 2 is to train a classifier, and the classifier learning branch applies a single linear layer $f_c(\cdot)$ to the image representation r to predict the class-wise logits, and cross-entropy loss is applied at this stage. Furthermore, we recommend a pipeline of diagnosis mechanism: Step 1 - the correlation of slice sequences (i.e., brain PET images) for each subject are analyzed, in the event that the majority of the slice sequences are predicted as an AD case, a diagnosis of positive AD is suggested. Otherwise, step 2 - the result of the clinical neuropsychological assessment is incorporated to conduct a joint diagnosis.

B. Anchor Points and Loss Function

An anchor point possesses only one positive sample, and all other samples are regarded as negative samples. Fig. 3(a) shows a typical self-supervised contrastive learning method in which an original graph is augmented by data to obtain two subsamples, which constitute the positive pairs between the subsamples and the negative pairs with the subsamples of other

data. Many fuzzy positive and fuzzy negative pairs may occur for each subsample in fuzzy supervised contrastive learning. In other words, in self-supervised contrastive (left), only one positive sample is obtained by cropping the image of anchor (i.e., an augmented version of the original PET slices image), and other images are used as negative samples of the batch. However, the fuzzy supervised contrastive (right), considered in this paper, contrasts the set of all samples from the same class as positives against the negatives from the remainder of the batch, while fuzzy smoothing the label information to reduce the rigidity of the labels. As shown in Fig. 3, taking class label information into account results in an embedding space where elements of the same class are more richer sample data than in the self-supervised case.

We carefully designed various ways of the data amplification, including image cropping and flipping which is a classical method for data augmentation, and the fuzzy amplification approach (such as image blurring, image gaussian blurring, image median blurring, image bilateral blurring) on the original data. Note that only about 10% of the total amount of images have been amplified using the fuzzy amplification approach. Furthermore, constructing a small set of samples using fuzzy amplification approach can help the network to improve the robustness and enhance the noise immunity of the network.

For the training data, although we selected PET slices that focused on the region in which the lesions occurred (the mid-axis section), there were still mismatches between the slice and label information, especially in some AD patients for whom some of the slices may still have been normal. To prevent network mislearning due to mislabeling and reduce the misdiagnosis rate of neuroimaging prediction, fuzzy mathematical theory [24] has been applied to the label generation processes of neuroimaging prediction networks as a way to reduce label rigidity and to counteract network mislearning caused by mislabeling via the supervised contrastive loss metric to measure the distances between different species, as shown in Fig. 3(b). The formula is as follows:

$$\hat{y}_i = y_i(1 - \alpha) + \alpha/\kappa \quad (1)$$

$$\hat{y}_i = \begin{cases} 1 - \alpha, i = \text{target} \\ \alpha/\kappa, i \neq \text{target} \end{cases} \quad (2)$$

Where, κ is the number of neural image prediction classification categories, α is 0.2, and y_i is the label of sample i .

In this paper, we proposed the fuzzy supervised contrastive loss (L_{FSC}) to fuzzy smooth the label information, which can reduce the label stiffness and improve the generalization ability of the network. To be more specific, in L_i^{self} , the positive image is an alternatively augmented view of the anchor image; on the contrary, in L_{FSC} , apart from the alternatively augmented counterpart, the positives also include some other images from the same class [13].

Therefore, we unified all fuzzy positive images of anchor point x_i as $\{x_i^+\} = \{x_j | \hat{y}_j = \hat{y}_i, i \neq j\}$ (we assumed that different views of the same image had different indices). The definitions of “fuzzy positive” and “fuzzy negative” for x_i also apply to Z_i , where positive is $\{z_i^+\}$ and negative is $\{z_i^-\}$. In general, assuming a small batch size N , the fuzzy supervised contrastive loss can be expressed as:

$$L_{FSC} = \sum_i^{2N} L_i^{\text{self}} \quad (3)$$

$$L_i^{\text{self}} = \frac{-1}{2N_{\hat{y}_i}} \sum_{j=1}^{2N} l_{i \neq j} \cdot \log \frac{\exp[(z_i \cdot z_{j(i)})/\tau]}{\sum_{k=1}^{2N} l_{i \neq k} \cdot \exp[(z_i \cdot z_k)/\tau]} \quad (4)$$

Where, $2N_{\hat{y}_i}$ is the number of positive samples of anchor point z_i and $\tau > 0$ is the scalar temperature parameter.

Compared to L_i^{self} , L_{FSC} allows more flexibility and smoother to combine any number of positive samples. It optimizes the agreement between these positive samples by comparing them to negative samples. However, computational complexity required for the method proposed in this paper is proportional to the total number of samples including positive and negative samples. This is undesirable under limited GPU computing conditions, and to reduce the resultant weakening of the network's representational learning ability, the classifier in this paper was trained separately by using the cross-entropy loss. The cross-entropy loss is formulated as follows:

$$L_{CE(\text{binarycross})} = \frac{1}{N} \sum_i [\hat{y}_i \cdot \log(p_i) + (1 - \hat{y}_i) \cdot \log(1 - p_i)] \quad (5)$$

$$L_{CE(\text{multi-class})} = -\frac{1}{N} \sum_i \sum_{c=1}^M y_{ic} \log(p_{ic}) \quad (6)$$

The purpose of using cross entropy is to train a classifier in order to optimize the network feature output and improve the classification performance of the network. The $L_{CE(\text{binarycross})}$ is used for training the classifier for binary classification, and the $L_{CE(\text{multi-class})}$ is used for training the classifier for triple classification. Where, N denotes the number of post-amplification training samples, M denotes the number of categories (2 or 3), y_i represents the label of sample i , p_i represents the probability that sample i is predicted to be a positive category, y_{ic} is the sign function (0 or 1), and p_{ic} is the predicted probability of sample i .

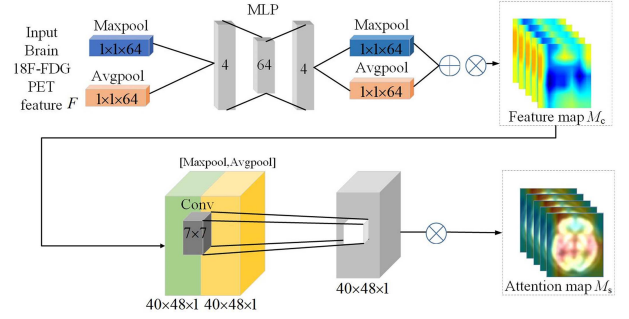


Fig. 4. Hybrid attention mechanism (HAM) for image features of slices.

C. Double Convolution Hybrid Attention Mechanism

Due to the clinical heterogeneity of individuals, different populations have different absorption strengths with respect to 18F-FDG imaging agents, causing the imaging results to vary. To solve this problem, we designed a double CNN network that can focus on more different regions and extract more features, as shown in Fig. 2.

Furthermore, to enable the model to learn the effective features of PET slices and suppress their redundant and invalid information, hybrid attention mechanism (HAM), a channel attention module and a spatial attention module, were designed as a way to enhance the effective feature learning ability of the network [25]. As shown in Fig. 4, firstly, a $40 \times 48 \times 64$ feature vector F is input, and then two $1 \times 1 \times 64$ channel vectors are obtained by the global average pooling and maximum pooling of feature F . These vectors are fed into a two-layer perceptual module with a first layer containing 4 neurons and an ReLU activation function and a second layer containing 64 neurons. Secondly, the two obtained features are output and summed after a sigmoid activation function to obtain the weight coefficients M_c , the features obtained by multiplying the weight coefficients with the original features F are fed into channel-dimensional average pooling and maximum pooling layers to obtain two $40 \times 48 \times 1$ channel vectors, and the two channel vectors are stitched together. A 7×7 convolution layer with a sigmoid activation function is applied to the feature, the weight coefficients M_s can then be obtained. Finally, the new features are obtained by multiplying the weight coefficients with the features F' . The formula is as follows:

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (7)$$

$$M_s(F) = \sigma(f^{7 \times 7}([AvgPool(M_c(F)), MaxPool(M_c(F))])) \quad (8)$$

Where, MLP is the two-layer perception module, $AvgPool$ is the average pooling module, $MaxPool$ is the maximum pooling module, F is the input feature vector, and f is the convolution module.

D. Multi-Correlation Analysis and Joint Diagnosis

Multi-correlation analysis is an indicator of the degree of correlation between a dependent variable and a set of independent variables (two or more). The prediction results obtained based on neuroimaging prediction networks can be classified into two

TABLE II
CLINICAL NEUROPSYCHOLOGICAL DIAGNOSTIC CRITERIA

	AD	MCI	NC
MMSE	27.0	10.0-27.0	<10.0
MMSEscore	0.0	1.0	2.0
CDR	0.0-0.5	0.5-1.0	1.0-3.0
CDRscore	0.0	1.0	2.0

or three categories. Where, 0 represents AD, 1 represents MCI, and 2 represents NC. If a subject's neuroimaging prediction results are consistent, the neuroimaging prediction results can be considered credible. If the neuroimaging predictions of the subject are inconsistent, the neuroimaging predictions are considered unreliable. In this paper, multi-correlation coefficients were used to calculate the correlations between the predicted outcomes of different brain 18F-FDG PET slices of subjects. Multi-correlation analysis was performed to determine whether the prediction network findings were consistent. The formula is calculated as follows:

$$R = \frac{\sum (y - \bar{y})(\hat{y} - \bar{\hat{y}})}{\sqrt{\sum (y - \bar{y})^2 (\hat{y} - \bar{\hat{y}})^2}} \quad (9)$$

Where, $\bar{y}$ is the mean of the predicted outcomes and y is the prediction for the firstly slice. $\hat{y}$ is the set of outcomes predicted for the 2nd to kth slices. The R values were between 0 and 1. The consistency of the neuroimaging prediction results was determined by their multi-correlation coefficients. When R is close to 1, the prediction is consistent, and the neuroimaging prediction result is credible. When R is close to 0, the neuroimaging prediction results are not credible.

We presents the following formula to combine the prediction network diagnosis with the clinical neuropsychological diagnosis result:

$$ADscore = \lambda \times avg \left(\sum_{i=1}^k PET_{cnn} \right) + (1 - \lambda) \times avg(MMSEscore + CDRscore) \quad (10)$$

Where, k is the PET slice of the subject and avg is the mean function. The MMSEscore is the score of MMSE and CDRscore is the score of CDR. ADscore is the result of a joint diagnosis combining prediction network with clinical neuropsychological assessment, as shown in Table II.

Therefore, the calculation formula for γ is as follows:

$$\gamma = R \quad (11)$$

Where, R is the multi-correlation coefficient. When $R = 1$, we takes the prediction network results. When $R = 0$, we only focus on the diagnosis of clinical neuropsychology.

V. EXPERIMENTS

In this section, we conducted a large number of experiments and analyses to comprehensively evaluate the performance of the proposed method.

A. Experimental Environment and Training Strategy

The model training and testing environment consisted of a Windows system, 6 GB of video memory, Python 3.8, the PyTorch framework, the Adam optimizer, a training learning rate of 0.0005, a maximum number of epochs of 100, a batch

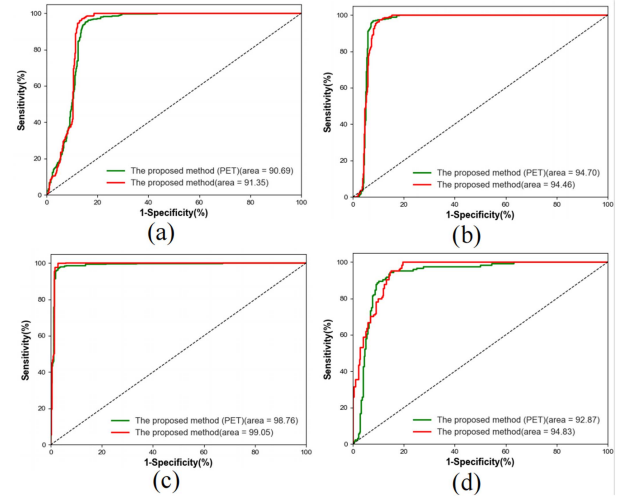


Fig. 5. ROC curves of the 2 AD prediction methods proposed in this paper in different classifications.

size of 64, and tenfold cross training. The mixed-precision training acceleration library was used, and the acceleration library parameters were set to accumulation-steps = 4 and opt-level = "O1". During the training process, the model was saved when the loss value was less than the optimal value.

B. Evaluation Metrics

In this paper, sensitivity is defined as the probability of correctly diagnosing a subject with AD/MCI. Specificity is the probability of correctly diagnosing the NC category. 1-Specificity is that a normal person is diagnosed as positive. The calculation formulas are as follows:

$$\begin{aligned} \text{Accuracy(in\%)} &= \frac{TP + TN}{TP + TN + FP + FN} \times 100 \\ \text{Sensitivity(in\%)} &= \frac{TP}{TP + FN} \times 100 \\ \text{Specificity(in\%)} &= \frac{TN}{TN + FP} \times 100 \end{aligned} \quad (12)$$

Where, TP are subjects with correct diagnosis of AD/MCI. FN are subjects with incorrect diagnosis of AD/MCI. FP are subjects with incorrect diagnosis of NC. TN is correctly diagnosed in subjects with NC by the prediction network.

The ROC curve is formed by sequential coordinate connections $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the AUC is as follows:

$$AUC = \frac{1}{2} \sum_{i=1}^{n-1} (x_{i+1} - x_i)(y_i + y_{i+1}) \quad (13)$$

C. Experimental Results and Analysis of Joint Diagnosis

Fig. 5 shows the ROC curves for the AD vs. NC, MCI vs. NC, AD vs. MCI, and AD vs. MCI vs. NC cases of the test set. From the ROC curves, it can be seen that the proposed method (the joint diagnostic) ROC curve proposed in this paper is closer to that of the clinician's diagnosis and has a larger AUC area value than the ROC curve of the proposed method (PET) (the single neuroimaging prediction network diagnosis approach). As shown in Fig. 5(d), the ROC curve of the combined diagnosis is

TABLE III
THE COMPARATIVE ANALYSIS OF THE VARIOUS INDICATORS USING THE PROPOSED METHOD AND OTHER DEEP LEARNING METHODS

Study	Subjects				Data		Classification Result(%)			
	AD	MCI	NC	Modality	Type	base	SEN	SPE	ACC	AUC
Pan et al. 2020 [28]	237	-	242	2D	PET	ADNI	90.32	95.49	93.13	97.11
Kim et al. 2020 [29]	141	-	348	2D	PET	ADNI	87.93	93.57	91.02	-
Zhang et al. 2019 [30]	91	-	101	2D	PET	ADNI	96.58	95.39	98.47	98.61
Liu et al. 2018 [31]	93	-	100	2D	PET	ADNI	91.4	91.0	91.2	95.3
Song et al. 2021 [26]	95	-	126	3D	PET	ADNI	89.13	94.27	92.10	-
Gao et al. 2021 [27]	196	-	227	3D	PET	ADNI	86.10	90.80	88.80	94.30
The proposed method (PET)	124	-	84	2D	PET	ADNI	97.18	96.29	98.54	98.76
The proposed method	124	-	84	2D+1D	PET+(MMSECDR)	ADNI	98.58	95.84	98.62	99.05
Liu et al. 2018 [31]	-	146	100	2D	PET	ADNI	78.10	80.00	78.90	83.90
Zhang et al. 2019 [30]	-	200	101	2D	PET	ADNI	84.54	75.36	84.62	87.44
Song et al. 2021 [26]	-	160	126	3D	PET	ADNI	72.81	70.56	72.00	-
The proposed method (PET)	-	192	84	2D	PET	ADNI	93.13	91.66	93.56	94.70
The proposed method	-	192	84	2D+1D	PET+(MMSECDR)	ADNI	91.57	90.11	91.26	94.46
Zhang et al. 2019 [30]	91	200	-	2D	PET	ADNI	94.97	79.24	84.20	87.92
Song et al. 2021 [26]	95	160	-	3D	PET	ADNI	57.46	89.69	78.30	-
The proposed method (PET)	124	192	-	2D	PET	ADNI	90.59	89.86	89.31	90.69
The proposed method	124	192	-	2D+1D	PET+(MMSECDR)	ADNI	90.24	89.17	90.42	91.35
Ding et al. 2018 [32]	36	79	73	2D	PET	ADNI	64.67	79	63.67	76.0
Song et al. 2021 [31]	95	160	126	3D	PET	ADNI	89.13	94.27	92.10	-
The proposed method (PET)	124	192	84	2D	PET	ADNI	91.18	88.23	91.93	92.87
The proposed method	124	192	84	2D+1D	PET+(MMSECDR)	ADNI	94.57	94.44	93.92	94.83

lower than the ROC curve of the single neuroimaging prediction network diagnosis when the 1-specificity is between 0-18% in the AD vs. MCI vs. NC classification cases because the level of susceptibility affects the results of the MMSE assessment in clinical neuropsychology, and the level of expertise of the diagnosing physician affects the results of the CDR assessment in clinical neuropsychology. Therefore, the key to diagnosing the accuracy of early AD results in subjects is the accuracy of neuroimaging predictions, this is consistent with a clinician's neurological disease diagnostic process, neurological disease diagnosis is a comprehensive clinical neuropsychological assessment aided by neuroimaging predictions.

Table III shows the proposed method in the AD vs. NC, MCI vs. NC, AD vs. MCI and AD vs. MCI vs. NC classification cases were compared with those of the other deep learning methods. This includes a 2D prediction method based on brain 18F-FDG PET slices and a 3D prediction method based on brain 18F-FDG PET slices. Among them, the methods of Song et al. [26] and Gao et al. [27] are 3D network-based AD prediction methods, and also some of the research hotspots because their networks learn the overall feature information of brain 18F-FDG PET slices and perform AD prediction with 3D networks. However, the drawback of this type of method is also obvious due to the presence of redundant information in brain 18F-FDG PET slices. Because a 3D network is computationally complex, it is not flexible for deep learning strategies with conventional computational resources, and the shortage of samples, which are needed to train the network, is one of the reasons why the 3D network prediction results are not as good as those of the slicing network, as shown in Table III. Therefore, the 3D network based on brain 18F-FDG PET for predicting AD diseases is limited.

When the slice-based AD prediction methods of Pan et al. [28], Kim et al. [29], Zhang et al. [30], Liu et al. [31] and Ding et al. [32] were compared to the 3D prediction network method, their prediction results were 5% higher overall in terms of the evaluation criteria and in the more difficult classification areas: The AD vs. MCI classification and the MCI vs. NC classification AUC values were 5.9% and 12.62% higher than those of the 3D network prediction results, respectively. This is because the advantages of brain 18F-FDG PET slice-based prediction are the availability of a large amount of training data, the flexibility to use deep strategies, and their low computational resource requirements. Considering clinical application requirements and computational resource limitations, 2D

prediction based on brain 18F-FDG PET slices was taken as the starting point of this paper.

Table III shows that both the single neuroimaging prediction model and the joint diagnosis method proposed in this paper outperformed the current methods. Specifically, in the AD vs. MCI cases and MCI vs. NC cases, traditional methods perform poorly in terms of classification accuracy; on the contrary, the proposed method (PET) in this paper stand out in the evaluation metric of AUC value that reaches 90.69% and 94.70%, respectively, which are significantly better (2.77% and 7.26%) than the predicted AUC values of the second-place method (Zhang et al. [30]).

We also observe from Table III that the joint diagnostic algorithm achieved an AUC value of 91.35% in the AD vs. MCI case, this value is better than the state-of-the-arts in terms of AUC value (specifically 3.43% improvement). However, it can be observed that the sensitivity and specificity of the single neuroimaging prediction model algorithm demonstrate better than that of the joint diagnostic algorithm. Because the proposed method (Joint) incorporates the neuropsychological diagnosis information such as the MMSE and CDR. However, the MMSE is subject to the level of education of individual. For instance, a patient of the MCI with high education level scores similar to a NC with low education level in the MMSE test. In such scenario, it is difficult to distinguish between MCI and NC. And the level of expertise of the diagnosing physician affects the results of the CDR assessment in clinical neuropsychology.

Additionally, in the AD vs. MCI vs. NC classification case, the misdiagnoses of MCI and AD were caused by label mismatches in the brain 18F-FDG PET slices (the AD slices may have contained MCI and NC features). Usually, AD predictions based on brain 18F-FDG PET slices are not as good as those of 3D networks, with the method in [26] yielding 28.43% higher ACC values than those of the method in [32]. This paper used a fuzzy supervised contrastive loss and a fuzzy cross-entropy loss to effectively reduce the amount of brain 18F-FDG PET slice misinformation caused by network mislearning, effectively obtain higher-level common features between different kinds of slices, and improve the generalization ability and classification performance of the prediction network. The ACC value of the single neuroimaging prediction model proposed in this paper was 92.87% on the test set, which was 28.66% higher than the ACC value of the traditional slice prediction network and 0.23% higher than the ACC value of the 3D prediction network.

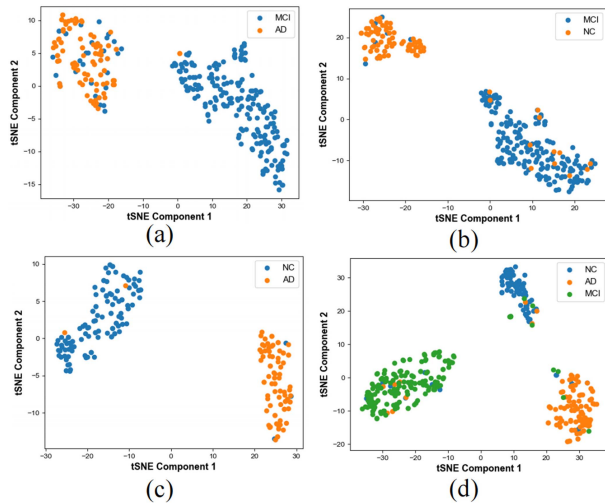


Fig. 6. Neuroimaging prediction results based on t-SNE visualization.

The ACC value of the joint diagnosis method on the test set was 30.25% higher than the ACC value of the traditional slice prediction network and 1.82% higher than the ACC value of the 3D prediction network. This indicates that the multiattention mechanism, anchor filter-based data augmentation method, the fuzzy supervised contrastive loss and the fuzzy cross-entropy loss are effective in optimizing the label information mismatches and representation learning ability of the developed network with respect to slices.

D. Neuroimaging Prediction Model Interpretation and Visualization

1) Performance Analysis of the Neuroimaging-Based AD Disease Prediction Model: The t-sne is a machine learning algorithm for data dimensionality reduction [33]. It can visualize the distribution of the network prediction results and test whether the network learns PET lesion features effectively. In this paper, we perform the output of feature results of PET slices for the first stage (feature learning stage) network, and process the feature results by t-sne method. Fig. 6(a) shows that MCI and AD are clustered more closely, and the mixed data lie between the two types, indicating that the neural image prediction model has difficulty in classifying AD and MCI. This is consistent with clinical diagnostic studies and generally makes it difficult for clinicians to determine whether an MCI subject already has AD from neuroimaging findings, reflecting the need for clinical neuropsychological assessment. Although the AD and MCI classes were poorly aggregated, they were largely separated, indicating that the neural image prediction network had good classification performance. In the AD vs. NC classification case in Fig. 6(c), almost all of the AD and NC samples were located in their own clusters, indicating that the network had high sensitivity to AD and NCs, furthermore, the dense and distant clustering effects achieves for the two classes in the MCI vs. NC classification case indicates that the prediction model already possessed discrimination ability, and this high classification performance is not possible with clinical neuropsychological assessment, thus, neuroimaging-based AD prediction is worthy of investigation. As shown in Fig. 6(d), the AD vs. MCI vs. NC classification test demonstrated that AD and NC samples were clustered more

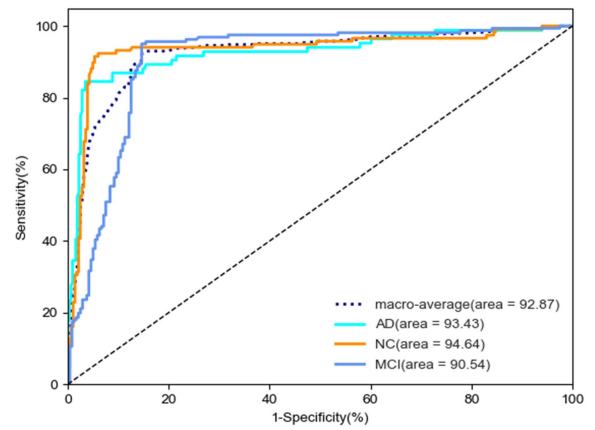


Fig. 7. The ROC curves of the single neuroimaging prediction network.

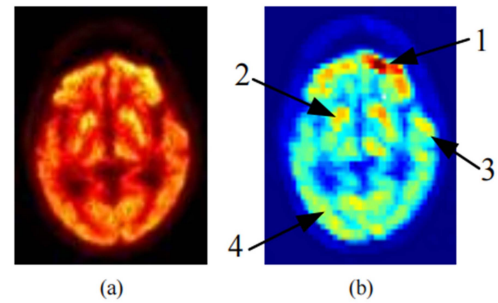


Fig. 8. The network attention areas for AD subjects.

densely, indicating that the prediction model obtained features between both classes. In addition, the MCI class was scattered, which was caused by the lack of distinctive features within the class, but the distribution of MCI subjects was still within a certain area, again indicating that the prediction model had the ability to distinguish MCI, and such a result is acceptable. In addition, there were a small number of MCI classes in the AD and NC clusters, which was caused by the inconspicuous features of the MCI class. This was also based on the limitation of neuroimaging prediction. Such misdiagnosis cases are common in the diagnosis process of clinical neurologists. Therefore, clinical neuropsychologist-assisted assessment is necessary.

In Fig. 7, the ROC curves of the single neuroimaging prediction network proposed (PET) are shown for the AD vs. MCI vs. NC classification case. Among them, the AUC values obtained with regard to the AD, MCI and NC were 93.43%, 90.54% and 94.64%, respectively. The ROC curves of the NC class were smooth, and the AUC values of the NC class were higher than those of the AD and MCI classes, indicating that the network had higher sensitivity and specificity for the NC class. Furthermore, the AUC value of the MCI class was the lowest, indicating that the model had difficulty identifying the MCI class. This is consistent with the results visualized in Fig. 6(d).

2) Model Interpretation: To explore the feature learning process in the neuroimage-based prediction network proposed (PET) in this paper, the model interpretability was investigated in depth. Experiments [32] were conducted to visualize the network layer features, as shown in Fig. 8 and 9.

Fig. 8(a) is a thermal map of the brain 18F-FDG PET slices of AD subjects, and Fig. 8(b) is a map of the regions of interest

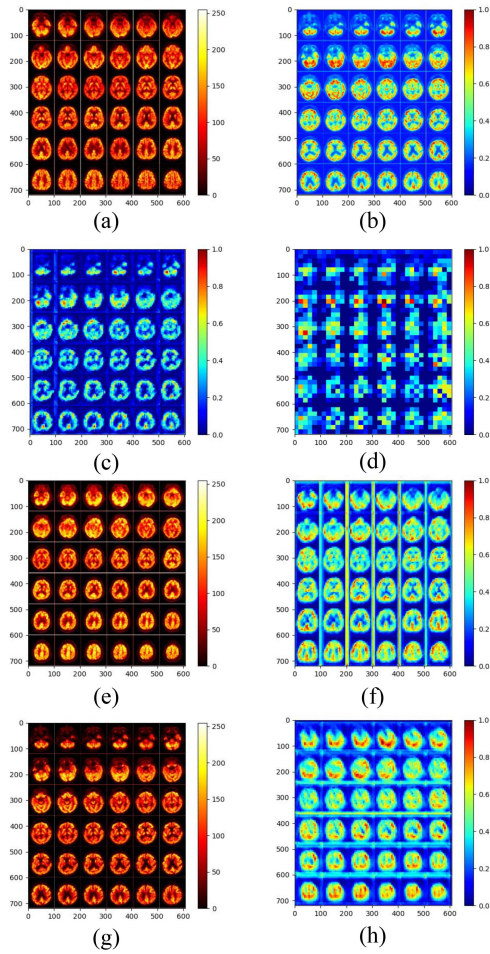


Fig. 9. The network attention areas for AD diagnosis.

determined by the neuroimaging prediction network proposed in this paper. This is the prediction network activated by weights to generate layer-specific feature map, which are then mapped to the range of 0-255 to generate ROI pseudo-color maps by the JET algorithm. In Fig. 8(b), label 1 is the right frontal region, label 2 is the left hippocampus region, label 3 is the right posterior gyrus region, and label 4 is the left occipital region of the AD subjects. This region of interest with the regions of AD lesions is consistent with diagnosis in clinical studies [34].

Fig. 9(a), (e), and (g) show the thermograms of the representative brain 18F-FDG PET slices of subjects in the AD, MCI and NC categories, respectively. Fig. 9(b) shows the region of interest in the brain 18F-FDG PET slice sequence images of the AD subjects after conv1. This region is not significantly different from that in Fig. 9(a) heatmap, but as the network layers deepen, as in Fig. 9(c), these sliceS feature are gradually refined from whole to local, and the focal AD features are extracted in the higher order network layer (Fig. 9(d)). These areas seem to be consistent with those that are most affected by AD, mainly in the hippocampus, posterior temporal lobe, parietal lobe, posterior cingulate gyrus and left parahippocampal gyrus. Fig. 9(f) shows the visualized feature images of the MCI subjects, and the MCI images focused on more regions than the visualized images of AD patients. They were especially sensitive to the bilateral postcentral gyrus regions, superior parietal regions, precuneus regions, left superior frontal gyrus regions, middle frontal gyrus

TABLE IV
DOUBLE CONVOLUTION HYBRID ATTENTION MODULE ABLATION EXPERIMENT

Method	AD vs. MCI vs. NC(%)			
	SEN	SPE	ACC	AUC
Without multiattention	86.33	83.24	87.65	88.34
Double convolution hybrid attention	91.18	88.23	91.93	92.87

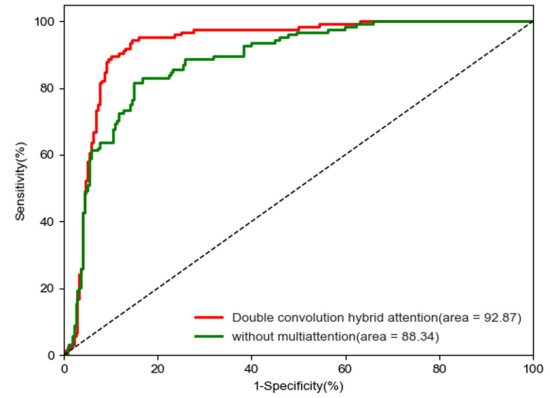


Fig. 10. ROC curve with/without double convolution hybrid attention.

regions and heel gyrus regions, this is consistent with the results of clinical neurological MCI disease studies. Fig. 9(h) shows the visualized images of the NC subjects' slices produced by the model. It can be found that the network is more sensitive to the occipital regions and temporal regions of NC slices, and whether these regions are possess lesions is also key to the detection of lesions in the subject's brain. The above analysis shows that the regions of interest yielded by the neuroimaging prediction model proposed in this paper are interpretable.

E. Neuroimaging Prediction Model Ablation Experiments

In order to verify the advancedness of the proposed method (PET) in this paper. The results of all ablation experiments were obtained in the AD vs. MCI vs. NC classification case.

1) *Double Convolution Hybrid Attention Module Ablation Experiment*: The results of the ablation experiments with respect to the multiattention module is shown in Table IV.

We trained the prediction network without the double convolution hybrid attention mechanism, and the other parameters were unchanged. The prediction results obtained on the test set were lower in terms of all metrics values than those of the double convolution hybrid attention prediction network. The AUC value of the prediction network containing the double convolution hybrid attention module was improved by 4.53%, which indicates that the double convolution and the HAM module are played active roles in the processes of focal feature learning and redundant information suppression.

Fig. 10 shows the ROC curves produced by the prediction network containing the double convolution hybrid attention module and the prediction network without the double convolution hybrid attention module on the test set. When the 1-specificity was between 0–10%, the ROC curve of the prediction network without the attention module was higher than that of the prediction network with the double convolution hybrid attention module, indicating that the network without multiattention was better for identifying NC classes. However, as the 1-specificity value became larger, the ability of the prediction network without

TABLE V

ABLATION EXPERIMENTS WITH DIFFERENT NETWORK STRUCTURES IN A SUPERVISED CONTRASTIVE LEARNING FRAMEWORK

Method	AD vs. MCI vs. NC(%)			
	SEN	SPE	ACC	AUC
AlexNet	78.25	77.48	79.56	79.15
VggNet-16	82.47	79.88	83.33	84.82
ResNet-34	90.83	87.26	91.89	92.31
DenseNet101	86.56	85.25	85.79	87.13
GooLeNetV3	81.23	80.32	84.66	84.82
EfficientNet-b0	77.08	73.26	77.13	78.09
ResNeSt-50	87.01	85.73	87.05	87.79
The proposed method (PET)	91.18	88.23	91.93	92.87

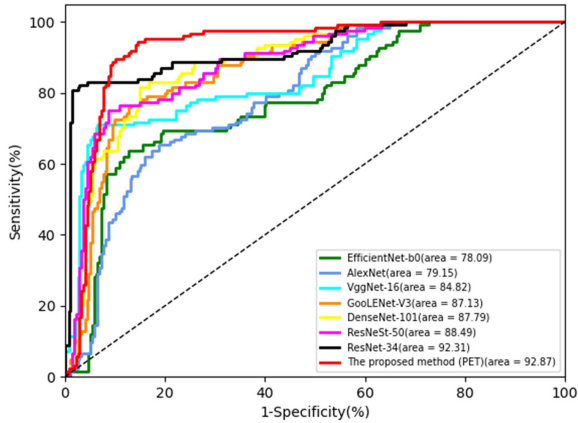


Fig. 11. ROC curve with different categories of modles.

the multiattention module to distinguish. The prediction network with the double convolution hybrid attention mechanism had a better ability to partition the AD and MCI samples. In clinical neurological disease diagnosis, the highest misdiagnosis rate occurred in the AD vs. MCI case, which was also unresolved by clinicians. Therefore, the double convolution hybrid attention network is more suitable for clinical neurological disease diagnosis.

2) Model Ablation Experiment: The ablation experiment results achieved by the proposed predictive network structure are shown in Table V.

We replaced several common networks with the same other conditions and verified the results on the test set. It can be found that the proposed method (PET) achieved the highest value in terms of each index and was followed by the residual network ResNet-34, which indicates that the jump connection of the residual module can suppress the redundant information contained in PET images and simultaneously fuse high-order feature information and low-order feature information to extract the focal information from the feature images. The poor test results of the ResNeSt101, GooLeNetV3 and DenseNet101 network structures were caused by the singular information of brain 18F-FDG PET slices. Network layers that were too deep caused the loss of important lesion information in the slices, and the redundancy of the slices made the prediction network learn more useless information, which led to a decrease in the AD prediction accuracy. Furthermore, we tested the EfficientNet structure with high classification performance, and its results on the test set were unfortunate. The reason for this is that EfficientNet relies heavily on pretraining weights. Without pretraining weights, its excellent scaling performance cannot be demonstrated, as shown in Fig. 11.

TABLE VI

ABLATION EXPERIMENTS AT DIFFERENT LOSSES

Method	AD vs. MCI vs. NC(%)			
	SEN	SPE	ACC	AUC
Cross-entropy loss	76.06	73.31	76.22	76.79
Self-supervised contrastive loss	80.12	79.59	80.91	84.82
Supervised contrastive loss	81.13	79.78	81.19	81.72
Fuzzy supervised contrastive loss	91.18	88.23	91.93	92.87

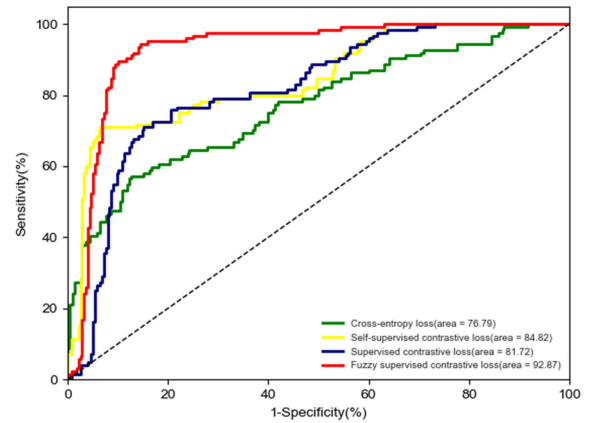


Fig. 12. ROC curve with different categories of loss.

3) Ablation Experiment Regarding the Loss Function: The results of the ablation experiments regarding the fuzzy supervised contrastive loss and fuzzy cross-entropy loss are shown in Table VI.

Three models, the traditional cross-entropy loss, self-supervised contrastive loss and supervised contrastive loss, were trained without changing the other parameters, and the test results of all three prediction models were lower than the test results of the fuzzy supervised contrastive loss proposed in this paper. Fig. 12 shows that the traditional cross-entropy loss prediction model and the supervised contrastive loss prediction model had higher AUC values when the 1-specificity was between 0-10% than that of the improved supervised contrastive loss prediction model proposed in this paper, which indicates that the cross-entropy loss and supervised contrastive loss are more sensitive to the NC class. The sensitivity of the self-supervised contrastive loss prediction model was higher than that of the traditional cross-entropy loss prediction model when the 1-specificity was greater than 10%, indicating that the self-supervised contrastive loss is more capable of distinguishing AD from MCI, which is related to its advantage of learning interclass image representation information. Thus, its performance improvement can be explained. Furthermore, when the 1-specificity reached 40%, the AUC value of the prediction model with excellent characterization learning performance and the supervised contrastive loss was lower than that of the self-supervised contrastive loss prediction model because the PET information of AD and MCI contained slices that do not match the label information. As a whole, the AUC value of the fuzzy supervised contrastive loss prediction model proposed in this paper was 8.05% higher than that of the second-ranked supervised contrastive loss prediction model. This is because the fuzzy supervised contrastive loss can effectively reduce the intensity of mislabeling and alleviate the mislabeling of slices (which would lead to model mislearning), making the classification curvature

TABLE VII
RESULTS OF DIFFERENT ANCHOR DATA AMPLIFICATION ABLATION EXPERIMENTS

Method	AD vs. MCI vs. NC(%)			
	SEN	SPE	ACC	AUC
Traditional anchor data augmentation	88.13	86.78	87.19	88.49
Optimized anchor data augmentation	91.18	88.23	91.93	92.87

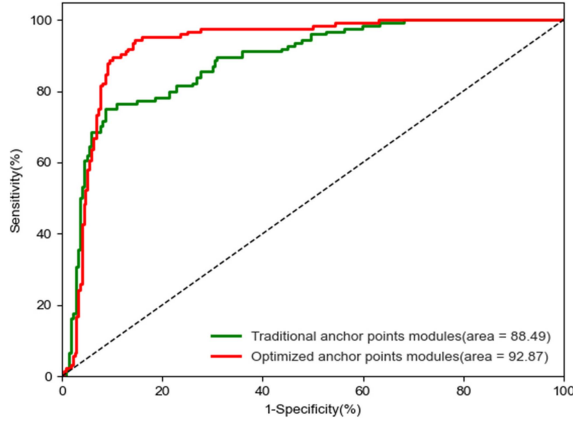


Fig. 13. ROC curve with different anchor data amplification methods.

of the network more reasonable. Additionally, the network was trained on ADNI-1 and tested on ADNI-2 to obtain high AUC values, indicating that fuzzy labeling is equally significant for improving the classification performance and robustness of the model.

4) Anchor Data Amplification-Based Ablation Experiment:

The results of the ablation experiments for the AD prediction model with the optimized anchor data augmentation method proposed in this paper are shown in Table VII.

The traditional anchor data augmentation-based prediction model was trained without changing the other parameters, and all of its prediction results on the test set were worse than those of the optimized anchor data augmentation prediction model. Fig. 13 shows that when the 1-specificity was between 0-10%, the AUC value of the traditional anchor point data amplification-based prediction model was higher than that of the optimized model, which indicates that the traditional anchor point data amplification-based prediction model is more suitable for identifying NC classes. When the 1-specificity was greater than 10%, the sensitivity of the improved anchor point amplification-based prediction model gradually became higher than that of the traditional anchor point data amplification model, which indicated that the improved anchor point data amplification-based prediction model was more sensitive to the AD and MCI classes, this was in accordance with our expected effect.

We further investigate the optimized anchor data augmentation (OADA) methods with noise (i.e., fuzzy amplification approach (such as image blurring, image gaussian blurring, image median blurring, image bilateral blurring) and traditional amplification approach (i.e., image cropping and flipping)) against traditional anchor data augmentation (TADA) methods (i.e., image cropping and flipping) and only original data methods (i.e., not anchor data augmentation (NADA)). From Fig. 14, we can find that the loss curves for the TADA method and the OADA method reach stability and fluctuations are decreased

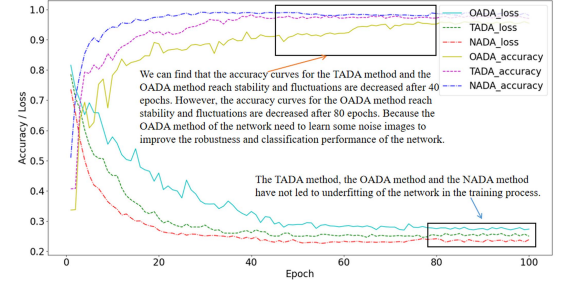


Fig. 14. Illustrate the loss curve (lower) and the accuracy curve (upper) plotted using different anchor data amplification methods.

TABLE VIII
RESULTS OF DIFFERENT DATA AMPLIFICATION METHODS LEAD NETWORKS OVERFITTING ABLATION EXPERIMENTS

Method	AD vs. MCI vs. NC(%)		
	Loss(Training)	Accuracy(Training)	Accuracy(Test)
OADA	23.91	97.15	82.35
TADA	25.04	96.49	87.19
OADA	27.79	95.48	91.93

after 70 epochs, indicating that the TADA method and the OADA method have not led to underfitting of the network.

Meanwhile, as shown in the Table VIII, the accuracy of the TADA method (96.49%) and the NADA method (97.15%) are better than that of the accuracy of the OADA method (95.48%) with the training data set, but with the testing data set, the accuracy of the TADA method (87.19%) and the NADA method (82.35%) are worse than that of the accuracy of the OADA method (91.93%), which indicates that the OADA method can improve the classification performance and the generalization ability of the network in the testing data set. It also can be seen that the prediction accuracy of the OADA method is 4.74% higher than the prediction accuracy of the TADA method and the OADA method is 9.58% higher than the prediction accuracy of the NADA method in the test. It indicates that the OADA method utilizing additional training samples, including constructing a small set of samples using fuzzy amplification approach can help the network to improve the robustness and classification performance, which can be confirmed by the experiment results.

VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed an AD prediction method based on contrastive learning, which alleviates several problems occurred in the existing methods: 1) the underperformance of traditional deep networks due to significant noise captured in PET images; 2) the unsatisfactory learning capability of existing deep networks for inter-class information and unreliable diagnostic results of a single method. Specifically, the novel contributions made in this paper can be summarized as follows: Firstly, the data of PET slices is augmented by cropping and blurring, and we introduced the label information of slice images including both positive and negative sample instances to generate more extended training data. We proposed the fuzzy supervised contrastive loss (i.e., L_{FSC}) extended from the contrastive loss function to reduce the large intra-class variation and expand inter-class variation simultaneously, which is one of the main challenges for PET images. Secondly, we constructed a double convolutional hybrid attention module, where two convolutional layers with different convolutional kernels (7×7 and 5×5) are

constructed to reduce the interference of low signal-to-noise for PET images, and enhance the ability of the network to learn different perceptual domains. In the experiment for AD prediction, the regions of interest extracted from the PET images using the proposed network are visualized using a heat map, and the results are consistent with those of the clinical study results, which indicates that the proposed method is effective in predicting AD. Furthermore, we recommended a pipeline of diagnosis mechanism: Step 1 - the correlation of slice sequences (i.e., brain PET images) for each subject are analyzed, in the event that the majority of the slice sequences are predicted as an AD case, a diagnosis of positive AD is suggested. Otherwise, step 2 - the result of the clinical neuropsychological assessment is incorporated to conduct a joint diagnosis.

In addition, many computer-aided diagnosis and risk assessment studies have been applied to the field of clinical neurological disease diagnosis. However, limited effort has been dedicated to the application of cutting-edge deep learning methods to clinical neuroimaging for disease prediction. Therefore, continuing to explore advanced deep learning algorithms for providing accurate early AD diagnoses to the global population is still a goal for both researchers and practitioners of the relevant fields in the future.

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